

HW1

Problem 1

```
slli t0,x29,2
add t0, x10,t0
lw x1,0(t0)
addi x1, x1, -5
```

```
slli t1,x28,2
add t1,x10,t1
lw x2,0(t1)
sub x2,x2,x1
sw x2, 32(x11)
```

Problem 2

(a)

- It's R-type, since its opcode is 0110011
 - 0000000_00001_00001_000_00001_0110011
 - add x1, x1, x1

(b)

- Instruction type: I-type, since its opcode is 0000011
- Assembly language instruction: li x3, -5(x27)
 - opcode = 0x3=0000011
 - rd=3=00011
 - funct3=0x0=000
 - rs1=27=11001
 - imm=-5=111111111011
 - binary :11111111011_11011_000_00011_0000011
- the hexadecimal representation of the instruction: 0xffbd8183

Problem 3

(a)

- first loop: x6=9, x5=2, 9!=0
- second loop: x6=8, x5=4, 8!=0
- third loop: x6=7, x5=6, 7!=0
- fourth loop: x6=6, x5=8, 6!=0
- x6=5, x5=10, 5!=0
- x6=4, x5=12, 4!=0
- x6=3, x5=14, 3!=0
- x6=2, x5=16, 2!=0
- x6=1, x5=18, 1!=0
- x6=0, x5=20, 0=0, done

Thus, the final value in register x5 is 20

(b)

```
#include <iostream>

int main(){
    int A = 0;
    int i = 10;
    while(i-->0){
        A += 2;
    }
}
```

Problem4

```
f:
    addi sp, sp, -12
    sw ra, 0(sp)
    sw a2, 4(sp)
    sw a3, 8(sp)

    jal g
    lw a3, 8(sp)
    lw a2, 4(sp)
    add a1, a2, a3
    jal g

    lw ra, 0(sp)
    addi sp, sp, 12
    ret
```

g:

```
add a0,a0,a1  
ret
```